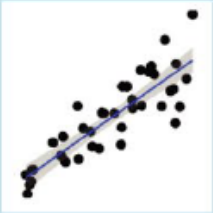
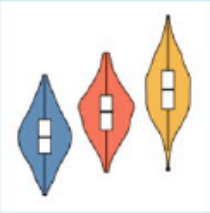
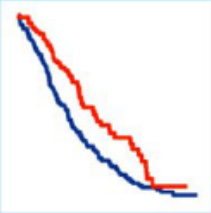
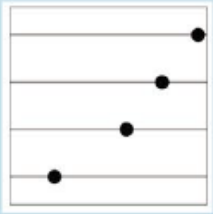
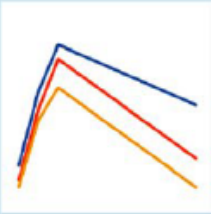
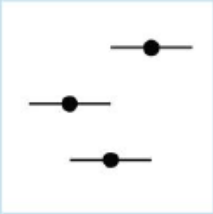
ARISTOTLE UNIVERSITY
OF THESSALONIKI

R data visualization

KONSTANTINOS I. BOUGIOUKAS, PhD

THESSALONIKI

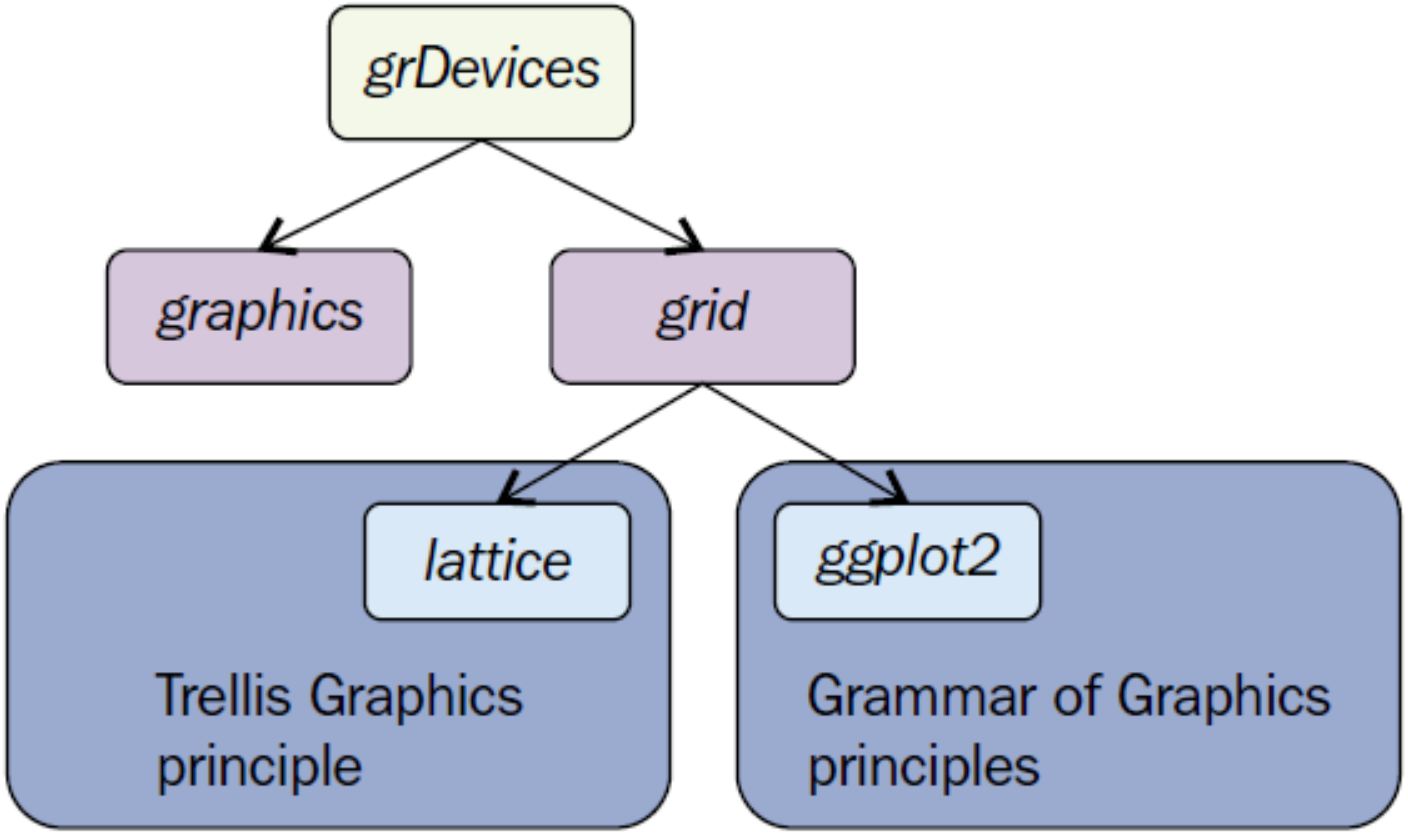
Chart types

Deviation	Correlation	Ranking	Distribution	Evolution	Part-to-whole	Magnitude
Chg. from baseline	Scatter plot	Horizontal bar chart	Boxplot	Kaplan Meier	Stacked bar chart	Vertical bar chart
						
Waterfall	Heat map	Dotplot	Histogram	Line plot	Tree map	Forest plot
						

Good data visualization

Step	Description	Notes
1.	Appropriate plot type for results	Might be a boxplot, a scatterplot, a linear regression fit ... many options
2.	Plot is well organised	The independent (explanatory) variable is on the x and the dependent (respnse) variable is on the y axis
3.	X and Y axes use correct units	Having proper symbols (for alpha, beta, etc.) and super/subscript where needed
4.	X and Y axes easy to read	Beware awkward fonts and tiny letters
5.	Clear informative legend	It's easy to tell apart what points/lines on the graph represent
6.	Plot is not cluttered	Don't put all results on one plot, give them space to shine
7.	Clear and consistent colour scheme	Stick with the same colours for the same variables, avoid red/green combinations which might look the same to colourblind people
8.	Plot is the right dimensions	Avoid overlapping labels and points/lines which merge together and make your graph longer/wider if needed
9.	Measures of uncertainty where appropriate	Error bars, confidence and credible intervals, remember to say in the caption what they are
10.	Concise and informative caption	Remember to include what the data points show (raw data? Model predictions?), what is the sample size for each treatment, the effect size and what measure of uncertainty accompanies it

The plotting environments in R



What is ggplot2 ?

- Based on the Grammar of Graphics, a coherent system for describing and building graphs.

- ggplot2 enables you to compose simple to complex graphs by combining independent components.

Basic elements to build a ggplot graph

Data

dataset

Aesthetics

color

position

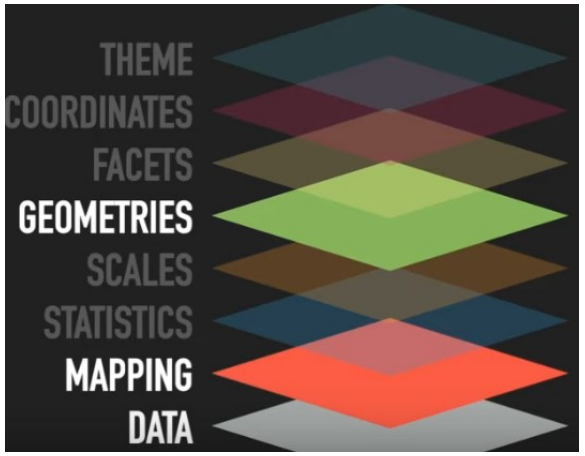
size

Geometry

points

lines

bars




```
ggplot(dataset, aes(x = var1, y = var2)) +
    geom_point()
```

Plot

`ggplot()`

- Initializes a ggplot object
- Adds **layers** to create graph
- No defaults, more control

ggplot() function

```
ggplot(dataset, aes(x = var1, y = var2)) + geom_point()
```

Data + aesthetics

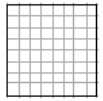
Add graph layer on top with **+**

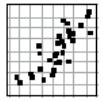
You can visualize the ggplot() function as drawing on a plain canvas. You start off with data, and then you add points and a coordinate system to get a final plot.

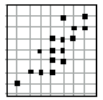
Geoms and their aesthetics

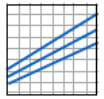
Each **geom** has a number of **aesthetics** that define its visual properties.

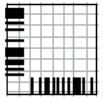
Continuous X, Continuous Y
f <- ggplot(mpg, aes(cty, hwy))


f + **geom_blank()**

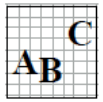

f + **geom_jitter()**
x, y, alpha, color, fill, shape, size

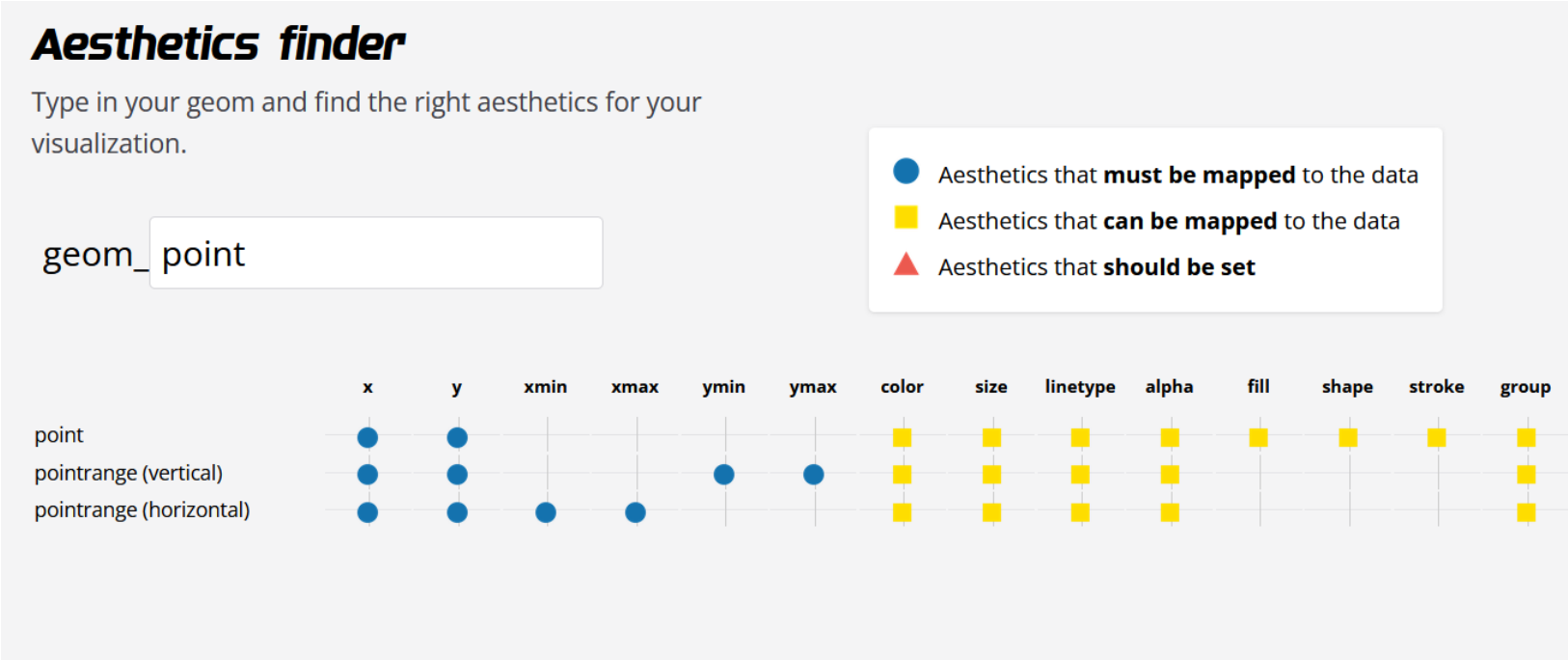

f + **geom_point()**
x, y, alpha, color, fill, shape, size


f + **geom_quantile()**
x, y, alpha, color, linetype, size, weight


f + **geom_rug**(sides = "bl")
alpha, color, linetype, size

f + **geom_smooth**(model = lm)
x, y, alpha, color, fill, linetype, size, weight

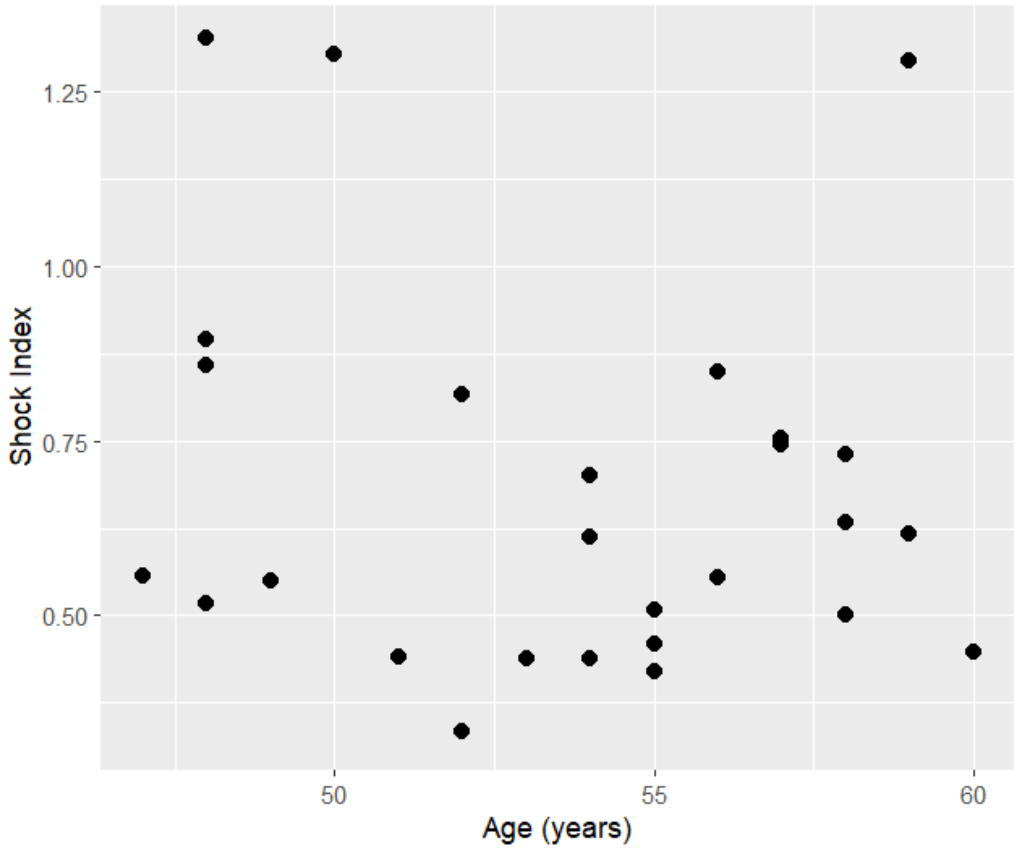

f + **geom_text**(aes(label = cty))
x, y, label, alpha, angle, color, family, fontface, hjust, lineheight, size, vjust



https://ggplot2tor.com/aesthetics

Example: scatter plot

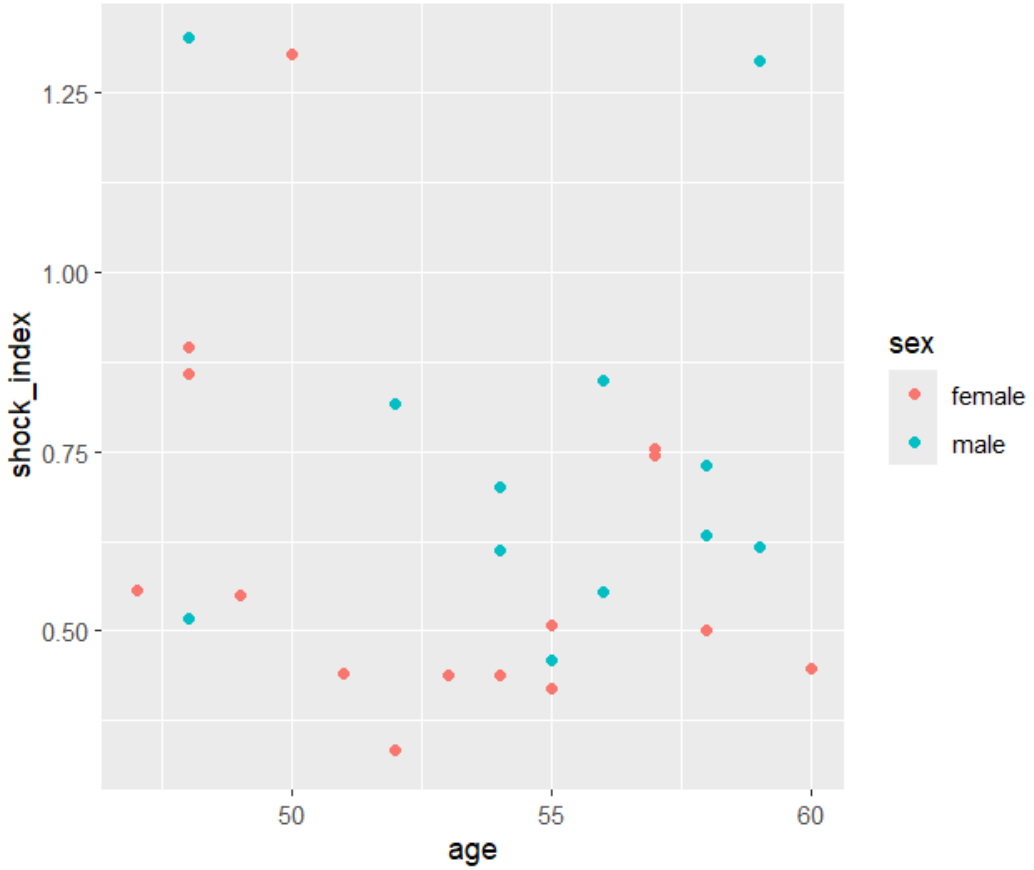
```
ggplot(data = dat10, mapping = aes(x = age, y = shock_index)) +
  geom_point()
```



In this case, geom_point() **inherits** the *x* and y aesthetics from the ggplot() function

Example: scatter plot- Color aesthetic

```
ggplot(dat10, aes(x = age, y = shock_index)) +
  geom_point(aes(color = sex))
```



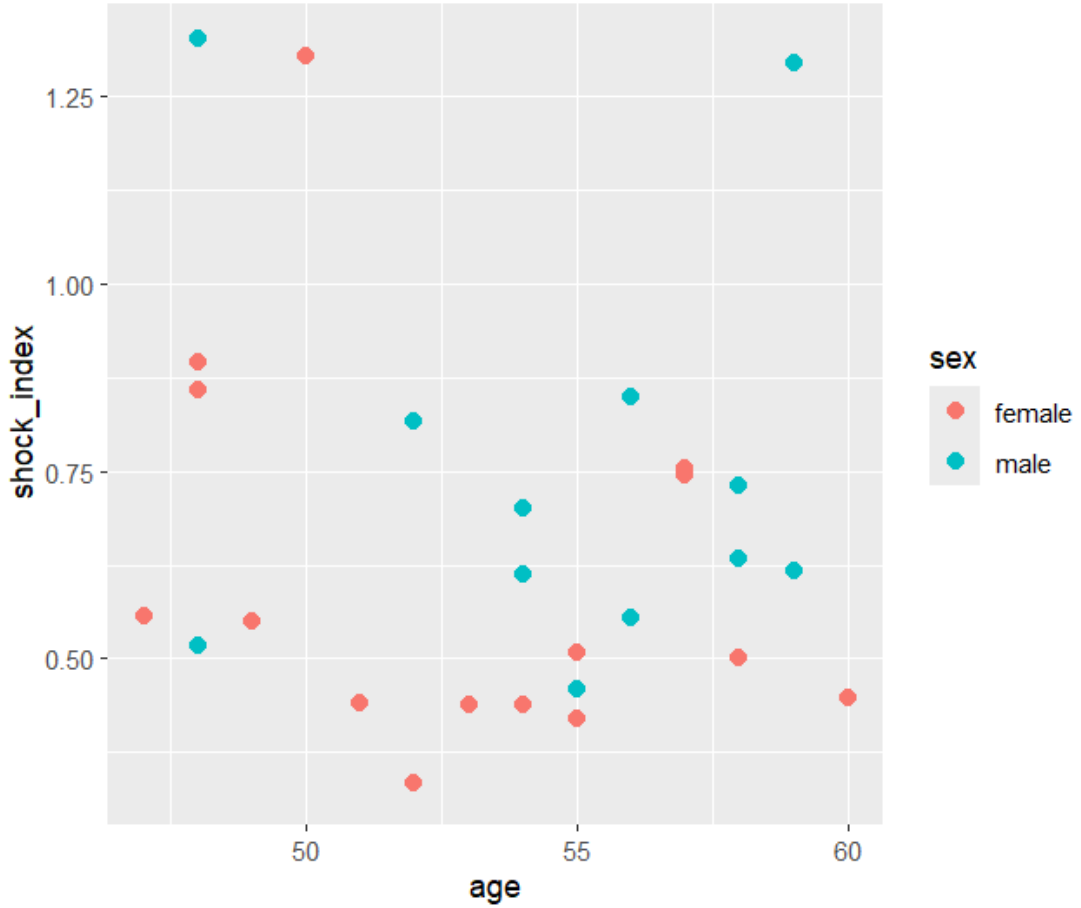
Here, we added **inside** the aes() of geom_point the **color** aesthetic.

The data of variable ***sex*** mapped to **color** aesthetic of geom_point.

Obviously, the qualitative scale was applied automatically by ggplot2

Example: scatter plot- Size aesthetic

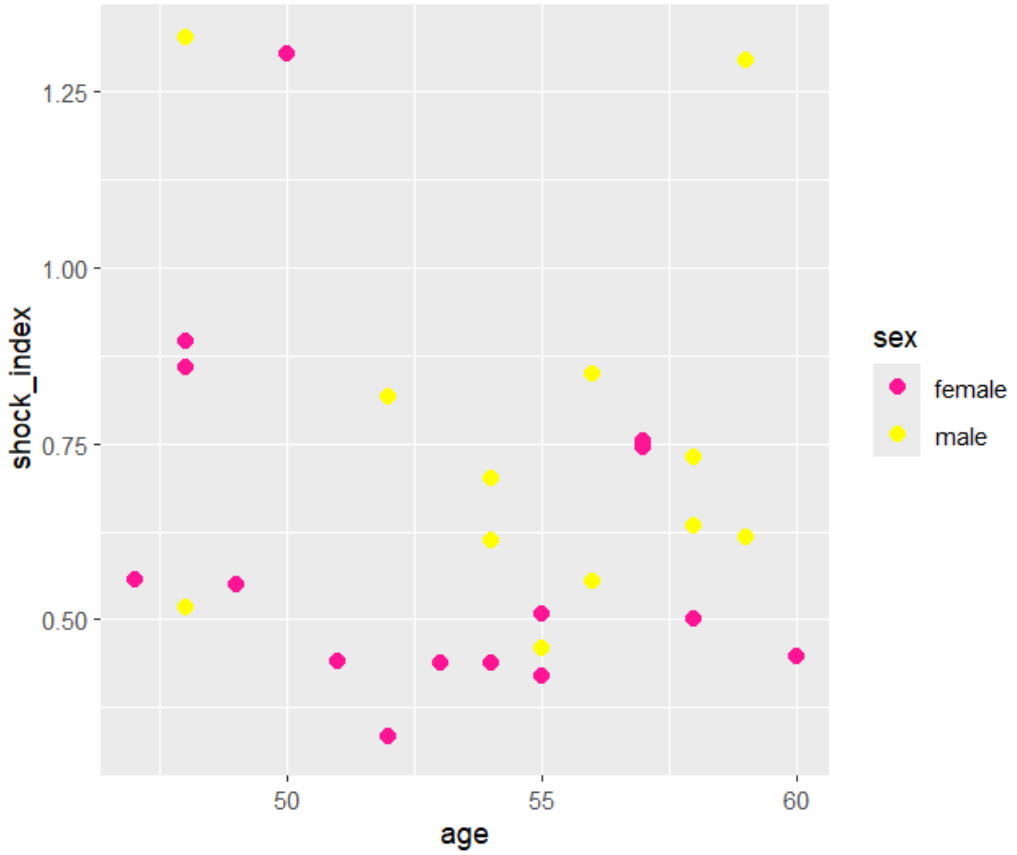
```
ggplot(dat10, aes(x = age, y = shock_index)) +
  geom_point(aes(color = sex), size = 2.5)
```



We can change the size of all the points using the **size** aesthetic outside the aes() function.

Example: scatter plot- scales

```
ggplot(dat10, aes(x = age, y = shock_index)) +
   geom_point(aes(color = sex), size = 2.5) +
   scale_color_manual(values = c("deeppink", "yellow"))
```



We can change the default colors of the points for both groups using the scale_color_manual function.

Theme elements

ggplot2 theme elements reference

Set `minimal` as the baseline theme:

```
theme_minimal() +
theme(theme.element = element_type())
```

Use `element_blank()` to **remove an element**

Axis titles, text, ticks, and **lines** can be specified per axis using theme inheritance by putting `.x/.y` at the end of the theme element.

`plot.title.position = "plot"`
`plot.caption.position = "plot"`
"plot" means that they will be aligned to the entire plot (instead of the panel)

`plot.title = element_text()`

`plot.subtitle = element_text()`

`plot.margin = margin(25, 25, 25, 25)`

`axis.line.y = element_line()`

`axis.title.y = element_text()`

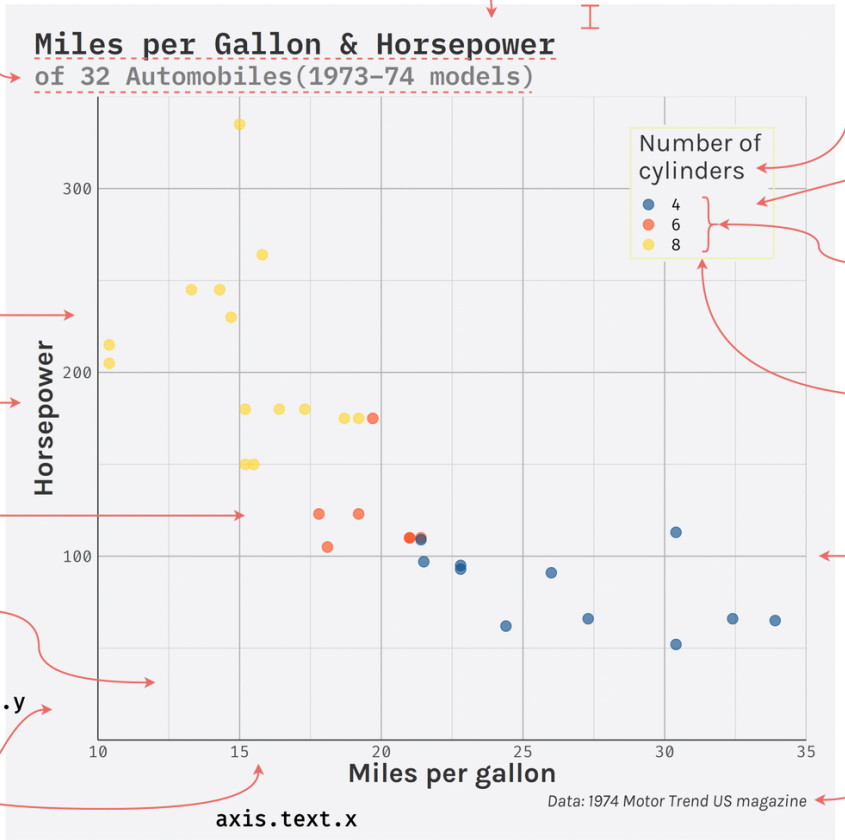
`panel.grid.major = element_line()`

`panel.grid.minor = element_line()`

`axis.text.y`

`axis.text = element_text()`

`axis.text.x`



`legend.title = element_text()`

`legend.background = element_rect()`

`legend.text = element_text()`

`legend.position = c(.85,.85) / "none"/ "left"/ "right"/ "bottom"/ "top"`

`plot.background = element_rect()`

`plot.caption = element_text()`

isabella-b

`text = element_text()` ← modifications will be applied to all text elements

Full list of elements at ggplot2.tidyverse.org/reference/theme

Themes can be used to give plots a consistent customized look to the **non-data elements** of the plot. Modify a single plot's theme using **theme()**